

Experiment no - 6

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AIM : To connect flutter UI with firebase database.

THEORY:

Introduction to Firebase and Flutter Integration

Firebase is a comprehensive platform developed by Google, designed to help developers build high-quality applications for both mobile and web. It provides essential services such as real-time databases, authentication, cloud storage, hosting, and much more. One of the most widely used Firebase services is the Firebase Realtime Database, which is a NoSQL cloud database that allows data to be stored and synced in real-time across all connected devices.

Flutter, on the other hand, is an open-source UI software development kit created by Google, which allows developers to build natively compiled applications for mobile, web, and desktop from a single codebase. Its rich set of pre-designed widgets and powerful tools makes Flutter an attractive option for developing visually appealing and performant applications. Integrating Firebase with Flutter allows developers to leverage the full potential of Firebase services in their applications. By using Firebase's Realtime Database, Flutter apps can achieve features such as real-time data synchronization, secure authentication, and cloud-based storage. This combination enables developers to create powerful, scalable, and feature-rich mobile and web applications.

Setting Up Firebase in Flutter :

To connect a Flutter app with Firebase, the following steps are typically followed:

1. **Creating a Firebase Project:** To start using Firebase with Flutter, the first step is to create a Firebase project in the Firebase Console. Once the project is created, developers can associate their Flutter app with the Firebase project by following the platform-specific instructions for Android or iOS. This usually involves configuring API keys, downloading configuration files, and adding them to the Flutter project.

2. **Integrating Firebase SDK in Flutter:** After the Firebase project is set up, developers need to integrate Firebase's SDK into the Flutter app. This involves adding the necessary dependencies to the Flutter project's pubspec.yaml file. For Firebase's Realtime Database, the package `firebase_database` is used.

Additionally, Firebase's core SDK (`firebase_core`) must also be included to initialize Firebase services.

3. **Initializing Firebase:** Before any Firebase functionality can be used, it is essential to initialize Firebase in the Flutter app. This is done by calling `Firebase.initializeApp()` in the main entry point of the app (usually in the `main.dart` file). Firebase needs to be initialized before interacting with any Firebase services, such as the Realtime Database, Cloud Firestore, or Authentication. Connecting Firebase to a Flutter app enables developers to create robust, scalable, and real-time applications with ease. Firebase's Realtime Database offers a powerful, cloud-based solution for managing data in realtime, while Firebase Authentication ensures secure access control. By integrating Firebase with Flutter, developers can take advantage of real-time data synchronization, offline support, and a wide range of other Firebase features, allowing them to build feature-rich apps that meet modern user expectations.

PROGRAM:

Signup.dart

```
import 'package:filmyapp/pages/bottomnav.dart';
import 'package:filmyapp/pages/login.dart';
import 'package:filmyapp/service/database.dart';
import 'package:filmyapp/service/shared_pref.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'package:flutter/material.dart';

class SignUp extends StatefulWidget {
  const SignUp({super.key});

  @override
  State<SignUp> createState() => _SignUpState();
}

class _SignUpState extends State<SignUp> {
  String email = "", password = "", name = "";
  TextEditingController namecontroller = TextEditingController();
  TextEditingController passwordcontroller = TextEditingController();
  TextEditingController mailcontroller = TextEditingController();

  registration() async {
    if (namecontroller.text != "" && mailcontroller.text != "") {
      try {
        UserCredential userCredential = await FirebaseAuth.instance
          .createUserWithEmailAndPassword(email: email, password: password);

        String uid = userCredential.user!.uid;

        Map<String, dynamic> userInfoMap = {
          "Name": namecontroller.text,
          "Email": mailcontroller.text,
          "Id": uid,
          "Image":
            "https://firebasestorage.googleapis.com/v0/b/barberapp-
            ebcc1.appspot.com/o/icon1.png?alt=media&token=0fad24a5-a01b-4d67-b4a0-676fbc75b34a",
        };

        await SharedPreferencesHelper().saveUserDisplayName(namecontroller.text);
        await SharedPreferencesHelper().saveUserEmail(mailcontroller.text);
        await SharedPreferencesHelper().saveUserId(uid); // Save UID here
        await SharedPreferencesHelper().saveUserImage(userInfoMap["Image"]);
        await DatabaseMethods().addUserDetails(userInfoMap, uid);

        ScaffoldMessenger.of(context).showSnackBar(
          const SnackBar(
```

```

        backgroundColor: Colors.green,
        content: Text(
          "Registered Successfully!",
          style: TextStyle(
            color: Colors.white,
            fontWeight: FontWeight.bold,
            fontSize: 20.0,
          ),
        ),
      ),
    ),
  );
  Navigator.pushReplacement(
    context, MaterialPageRoute(builder: (context) => BottomNav()));
} on FirebaseAuthException catch (e) {
  String errorMessage = e.code == 'weak-password'
    ? "Password Provided is too Weak"
    : "Account Already exists";

  ScaffoldMessenger.of(context).showSnackBar(
    SnackBar(
      backgroundColor: Colors.orangeAccent,
      content: Text(
        errorMessage,
        style: const TextStyle(fontSize: 18.0),
      ),
    ),
  );
}
}
}

```

```

@override
Widget build(BuildContext context) {
  return Scaffold(
    backgroundColor: Color(0xff14141d),
    body: SingleChildScrollView(
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          Image.asset("images/signin.png", height: 200, fit: BoxFit.cover),
          Padding(
            padding:
              const EdgeInsets.symmetric(horizontal: 30.0, vertical: 40.0),
            child: Column(
              crossAxisAlignment: CrossAxisAlignment.start,
              children: [
                const Text(
                  "Sign Up",
                  style: TextStyle(
                    color: Colors.white,
                    fontSize: 32.0,

```

```

        fontWeight: FontWeight.bold,
      ),
    ),
    const SizedBox(height: 20.0),
    const Text("Name",
      style: TextStyle(color: Colors.white, fontSize: 18.0)),
    TextField(
      controller: namecontroller,
      decoration: const InputDecoration(
        hintText: "Enter Name",
        hintStyle: TextStyle(color: Colors.white54),
        suffixIcon: Icon(Icons.person, color: Colors.white),
      ),
      style: const TextStyle(color: Colors.white),
      cursorColor: Colors.white,
    ),
    const SizedBox(height: 20.0),
    const Text("Email",
      style: TextStyle(color: Colors.white, fontSize: 18.0)),
    TextField(
      controller: mailcontroller,
      decoration: const InputDecoration(
        hintText: "Enter Email",
        hintStyle: TextStyle(color: Colors.white54),
        suffixIcon: Icon(Icons.email, color: Colors.white),
      ),
      style: const TextStyle(color: Colors.white),
      cursorColor: Colors.white,
    ),
    const SizedBox(height: 20.0),
    const Text("Password",
      style: TextStyle(color: Colors.white, fontSize: 18.0)),
    TextField(
      controller: passwordcontroller,
      obscureText: true,
      decoration: const InputDecoration(
        hintText: "Enter Password",
        hintStyle: TextStyle(color: Colors.white54),
        suffixIcon: Icon(Icons.lock, color: Colors.white),
      ),
      style: const TextStyle(color: Colors.white),
      cursorColor: Colors.white,
    ),
    const SizedBox(height: 30.0),
    Center(
      child: GestureDetector(
        onTap: () {
          if (namecontroller.text != "" &&
            mailcontroller.text != "" &&
            passwordcontroller.text != "") {
            setState() {

```

```

        name = namecontroller.text;
        email = mailcontroller.text;
        password = passwordcontroller.text;
    });
    registration();
  }
},
child: Container(
  width: 160,
  padding: const EdgeInsets.all(12),
  decoration: BoxDecoration(
    color: Color(0xff6b63ff),
    borderRadius: BorderRadius.circular(30),
  ),
  child: const Center(
    child: Text(
      "Sign Up",
      style: TextStyle(
        color: Colors.white,
        fontSize: 20.0,
        fontWeight: FontWeight.bold,
      ),
    ),
  ),
),
),
),
),
),
const SizedBox(height: 20.0),
Center(
  child: GestureDetector(
    onTap: () => Navigator.push(
      context,
      MaterialPageRoute(builder: (context) => const LogIn()),
    ),
  child: const Text(
    "Already have an account? Log In",
    style:
      TextStyle(color: Color(0xff6b63ff), fontSize: 16.0),
  ),
),
),
),
),
),
),
),
),
);
}
}

```

Login.dart

```
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:filmyapp/pages/bottomnav.dart';
import 'package:filmyapp/pages/signup.dart';
import 'package:filmyapp/service/shared_pref.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'package:flutter/material.dart';

class LogIn extends StatefulWidget {
  const LogIn({super.key});

  @override
  State<LogIn> createState() => _LogInState();
}

class _LogInState extends State<LogIn> {
  String email = "", password = "";
  TextEditingController passwordcontroller = TextEditingController();
  TextEditingController mailcontroller = TextEditingController();

  userLogin() async {
    try {
      UserCredential userCredential = await FirebaseAuth.instance
        .signInWithEmailAndPassword(email: email, password: password);

      // Fetch user data from Firestore
      var userDoc = await FirebaseFirestore.instance
        .collection("users")
        .doc(userCredential.user!.uid)
        .get();

      if (userDoc.exists) {
        await SharedPreferencesHelper().saveUserId(userCredential.user!.uid);
        await SharedPreferencesHelper().saveUserDisplayName(userDoc['Name']);
        await SharedPreferencesHelper().saveUserEmail(userDoc['Email']);
        await SharedPreferencesHelper().saveUserImage(userDoc['Image']);
      } else {
        print("User document not found in Firestore.");
      }
    }

    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(
        content: Text("Logged in successfully!", style: TextStyle(color: Colors.white)),
        backgroundColor: Colors.green,
        duration: Duration(seconds: 2),
      ),
    );

    Navigator.push(context, MaterialPageRoute(builder: (context) => BottomNav()));
  } on FirebaseAuthException catch (e) {
```

```
String errorMessage = e.code == 'user-not-found'  
  ? "No user found for that email."  
  : "Wrong password provided.";
```

```
ScaffoldMessenger.of(context).showSnackBar(  
  SnackBar(  
    backgroundColor: Colors.white,  
    content: Text(errorMessage, style: TextStyle(fontSize: 16.0, color: Colors.black)),  
  ),  
);  
}  
}
```

@override

```
Widget build(BuildContext context) {  
  return Scaffold(  
    backgroundColor: Color(0xff14141d),  
    body: SingleChildScrollView(  
      child: Column(  
        crossAxisAlignment: CrossAxisAlignment.start,  
        children: [  
          Image.asset("images/signin.png", height: 200, fit: BoxFit.cover),  
          Padding(  
            padding:  
              const EdgeInsets.symmetric(horizontal: 30.0, vertical: 30.0),  
            child: Column(  
              crossAxisAlignment: CrossAxisAlignment.start,  
              children: [  
                const Text(  
                  "Log In",  
                  style: TextStyle(  
                    color: Colors.white,  
                    fontSize: 32.0,  
                    fontWeight: FontWeight.bold,  
                  ),  
                ),  
                const SizedBox(height: 40.0),  
                const Text(  
                  "Email",  
                  style: TextStyle(color: Colors.white, fontSize: 18.0),  
                ),  
                TextField(  
                  controller: mailcontroller,  
                  decoration: const InputDecoration(  
                    hintText: "Enter Email",  
                    hintStyle: TextStyle(color: Colors.white54),  
                    suffixIcon: Icon(Icons.email, color: Colors.white),  
                  ),  
                  style: const TextStyle(color: Colors.white),  
                  cursorColor: Colors.white,  
                ),  
              ],  
            ),  
          ],  
        ),  
      ),  
    ),  
  );  
}
```

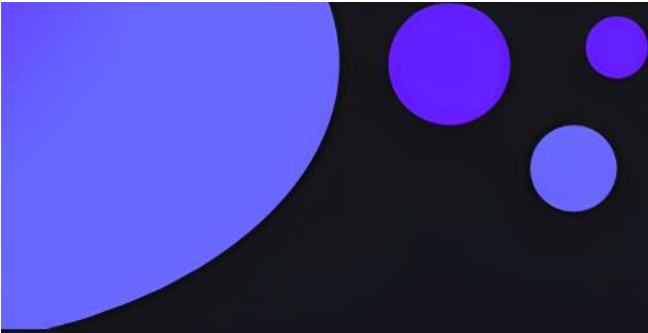
```

const SizedBox(height: 40.0),
const Text(
  "Password",
  style: TextStyle(color: Colors.white, fontSize: 18.0),
),
TextField(
  controller: passwordcontroller,
  obscureText: true,
  decoration: const InputDecoration(
    hintText: "Enter Password",
    hintStyle: TextStyle(color: Colors.white54),
    suffixIcon: Icon(Icons.lock, color: Colors.white),
  ),
  style: const TextStyle(color: Colors.white),
  cursorColor: Colors.white,
),
const SizedBox(height: 40.0),
Center(
  child: GestureDetector(
    onTap: () {
      if (mailcontroller.text != "" &&
        passwordcontroller.text != "") {
        setState(() {
          email = mailcontroller.text;
          password = passwordcontroller.text;
        });
      }
      userLogin();
    },
    child: Container(
      width: 160,
      padding: const EdgeInsets.all(12),
      decoration: BoxDecoration(
        color: Color(0xff6b63ff),
        borderRadius: BorderRadius.circular(30),
      ),
      child: const Center(
        child: Text(
          "Log In",
          style: TextStyle(
            color: Colors.white,
            fontSize: 20.0,
            fontWeight: FontWeight.bold,
          ),
        ),
      ),
    ),
  ),
),
const SizedBox(height: 20.0),
Center(

```



```
child: GestureDetector(  
  onTap: () => Navigator.push(  
    context,  
    MaterialPageRoute(builder: (context) => const SignUp()),  
  ),  
child: const Text(  
  "New user? Sign Up",  
  style:  
    TextStyle(color: Color(0xff6b63ff), fontSize: 16.0),  
),  
,  
,  
,  
  ],  
,  
,  
  ],  
,  
,  
);  
}  
}
```



Log In

Email

Enter Email



Password

Enter Password



Log In

New user? [Sign Up](#)

Sign Up

Name

Enter Name



Email

Enter Email



Password

Enter Password



Sign Up

Already have an account? [Log In](#)